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SYSTEM ANALYSIS

Old School Ice Cream Shop

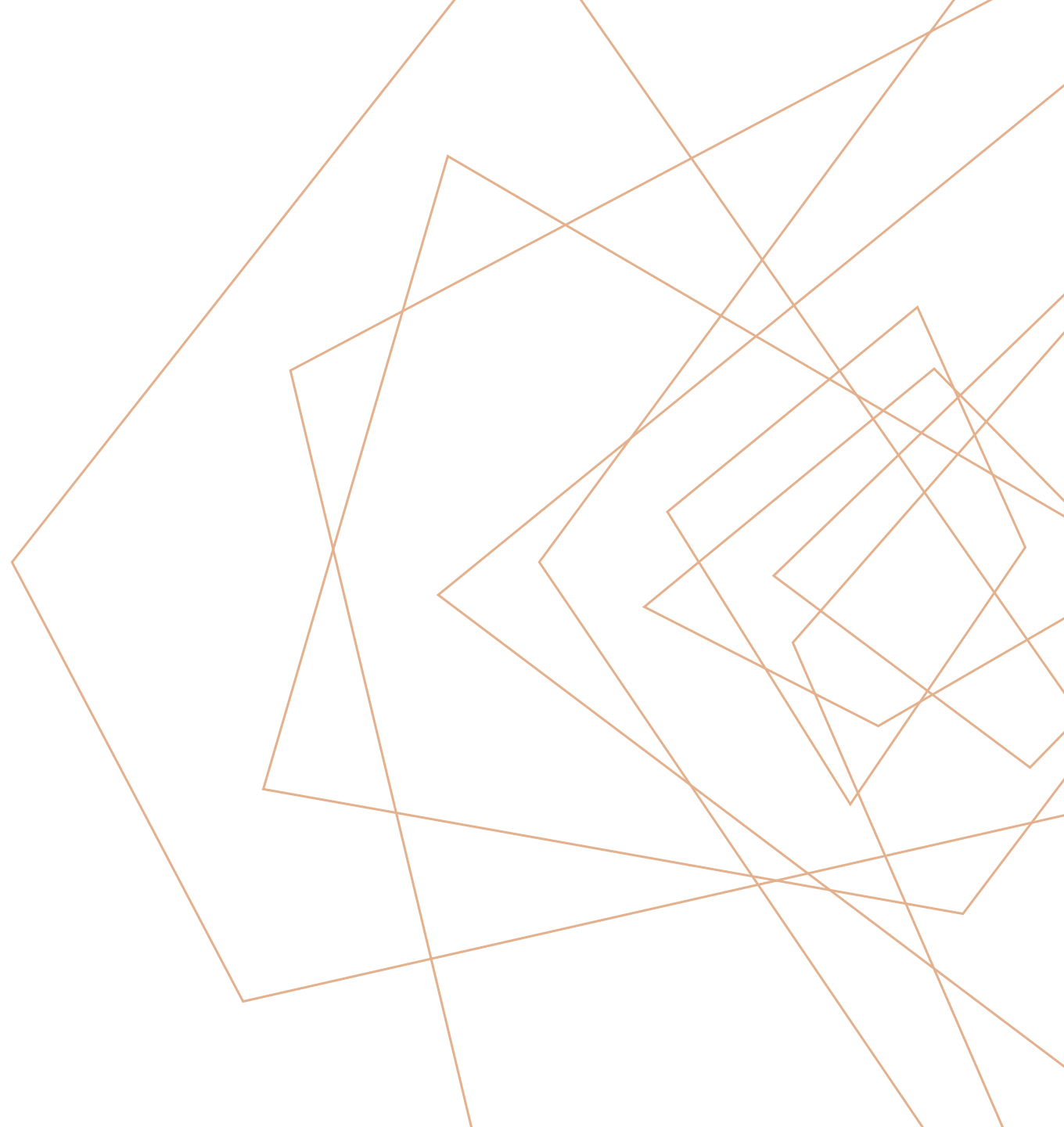
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PORTFOLIO CASE STUDY

Designed to demonstrate systems analysis and project planning skills using the SDLC framework.

MODERNIZING OPERATIONS FOR A RETAIL POS SYSTEM

The Old School Ice Cream Shop is facing declining sales due to outdated, manual operations. This case study proposes a modern digital solution designed to drastically improve efficiency, inventory tracking, and customer engagement. Using the SDLC framework, the following proposal outlines the system analysis, requirements, models, and implementation plan.



SCOPE OF WORK

SYSTEM DEVELOPMENT APPROACH

Analyze the current business problems and apply SDLC phases to design a modernized system.

SYSTEM CHARACTERISTICS ANALYSIS

Identify key system characteristics to understand how the new system should function.

PROCESS BREAKDOWN & REQUIREMENTS

Break the system into customer facing processes and define clear requirements for each part.

VISUAL MODELING & DATA DESIGN

Create visual models of the system using Data Flow Diagrams and an ERD to show how data and processes connect.

SDLC PHASES

(APPLIED TO OLD SCHOOL ICE CREAM SHOP)

Planning

- Need for a new or enhanced system is identified.
 - Needs are identified, analyzed, prioritized, and arranged.
 - Scope of the proposed system is determined.
- A baseline project plan is developed.

Analysis

- System requirements are gathered from user input and structured.
 - Current manual and computerized processes are carefully studied.
- Problems and opportunities are identified.
 - Output is a description of the recommended alternative solution.

SDLC PHASES CONT.

(APPLIED TO OLD SCHOOL ICE CREAM SHOP)

Analysis

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Implementation

- System is coded, tested, installed, and supported in the organization.
 - New system becomes part of daily operations.
- User training and documentation are completed.

SYSTEM CHARACTERISTICS 1-5

1. **Components** : The ice cream shop system is made of parts like the cash register screen, the inventory tracker, the customer loyalty records, reports, and the main database. These parts together form the system.
2. **Interrelated Components** : The parts of the system work together. For example, when the cashier records a sale, the inventory tracker and the customer loyalty records update at the same time.
3. **Purpose** : The purpose of the system is to help the shop run better. It records sales, tracks stock and helps keep customers coming back.
4. **Boundaries** : The physical boundary is the computer, tablet, receipt printer, and other devices in the shop where the system runs. The logical boundary is the software and data you can reach through the system, such as sales records and customer accounts.
5. **Interfaces** : Interfaces are where the system meets other things. In this shop that includes the cashier screen, the manager screen, the card reader, and the receipt printer.

SYSTEM CHARACTERISTICS 6-9

6. **Constraints** : Constraints are limits of the system. For this shop they include the budget for equipment, the internet and power in the store, and the time staff must use the system correctly.
7. **Environment** : The environment is everything around the system that can affect it. For this shop that includes the store building, the network, backup storage, and safety rules for handling payment and customer data.
8. **Input** : Input is what goes into the system. Examples are customer orders, flavor and price changes, new stock counts, and staff logins.
9. **Output** : Output is what comes out of the system. Examples are printed receipts, daily sales totals, inventory alerts, and loyalty point balances for customers.

CUSTOMER FACING EVENTS

- Walk into the shop
- Look at menu board
- **Select ice cream flavor or item**
- **Cashier enters order into the system**
- **Customer pays with preferred form of payment**
 - System processes payment
- **Loyalty account is checked or created**
 - **Loyalty points are added**
- Wait while order is prepared
 - Receive ice cream
- **Receipt printed if needed**
 - Leave the shop

COMPONENTS AND REQUIREMENTS

1. Take Order:

- A. The system shall allow staff to add or select ice cream items from a menu.
- B. The system shall show the current order summary and total price.
- C. The system shall update the order when the staff makes changes.

2. Make Payment:

- A. The system shall support multiple payment methods.
- B. The system shall process the payment provided by the customer.
- C. The system shall show payment confirmation on the screen.

3. Apply Loyalty:

- A. The system shall look up a customer loyalty account or allow staff to create one.
- B. The system shall add loyalty points to the customer after purchase.
- C. The system shall show the updated point balance to the staff.

4. Update Inventory:

- A. The system shall reduce inventory amounts when an order is completed.
- B. The system shall alert staff when a flavor or topping is running low.
- C. The system shall allow staff to update inventory counts manually if needed.

5. Print Receipt:

- A. The system shall provide a printed or digital receipt for the customer.
- B. The system shall show order details and payment information on the receipt.
- C. The system shall include loyalty points earned on the receipt.

6. Prepare Order:

- A. The system shall send the confirmed order to the preparation area.
- B. The system shall show the list of items needing preparation.
- C. The system shall update the preparation status when staff marks it done.

7. Complete Order:

- A. The system shall allow staff to verify the order before giving it to the customer.
- B. The system shall mark the order as completed once handed off.
- C. The system shall store the order in the daily sales log.

DATA FLOW DIAGRAM WRITE UP

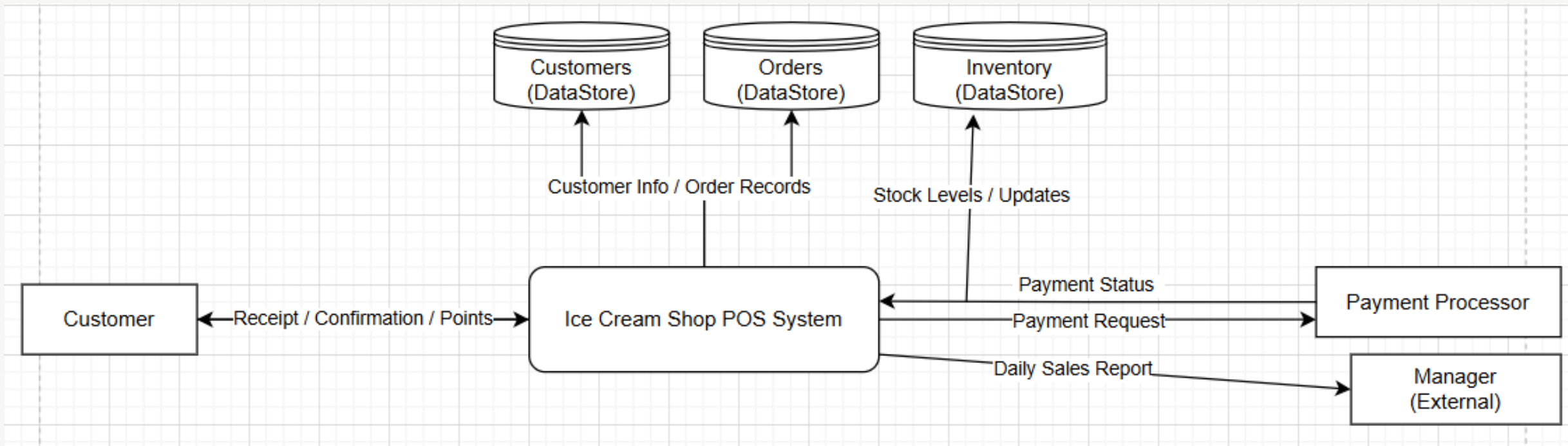
Processes – Take Order; Apply Loyalty; Make Payment; Update Inventory; Complete Order

Source – Customer

Sink – Payment Processor; Store Manager

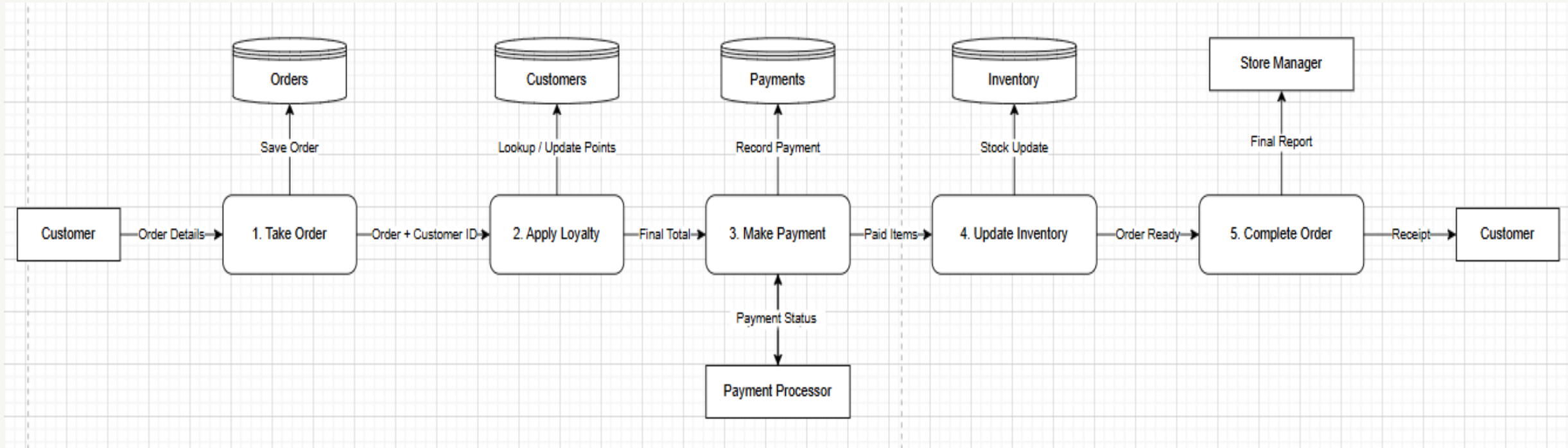
Data Stores – Customers; Orders; Payments; Inventory

CONTEXT DIAGRAM



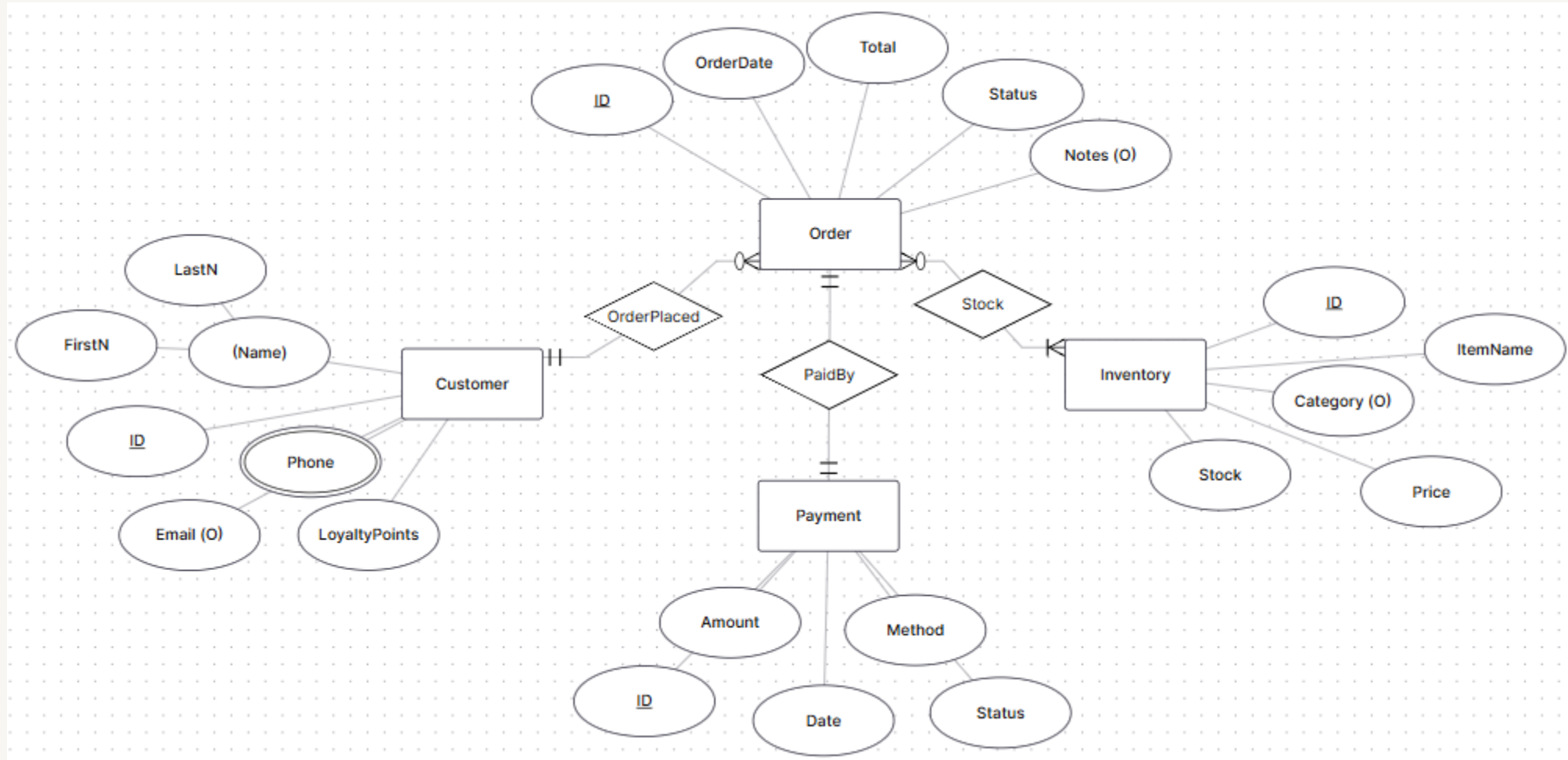
This diagram shows the system as one main process and the information it receives and sends. It also shows the customer, payment processor, and manager as the outside groups connected to the system.

LEVEL-0 DIAGRAM



This diagram breaks the system into smaller steps. It shows how the order is taken, how loyalty points and payments are handled, how inventory is updated, and how the order is finished. It also shows which data stores each step uses.

ERD DIAGRAM



This diagram shows the main things the system keeps track of, like customers, orders, payments, and inventory. It also shows the important details/attributes for each one and their relationships.

PROJECT SCHEDULE OVERVIEW

This schedule outlines all project activities, their durations, and dependencies.

The Gantt chart visually maps the full project timeline from planning to installation.

Gantt Chart

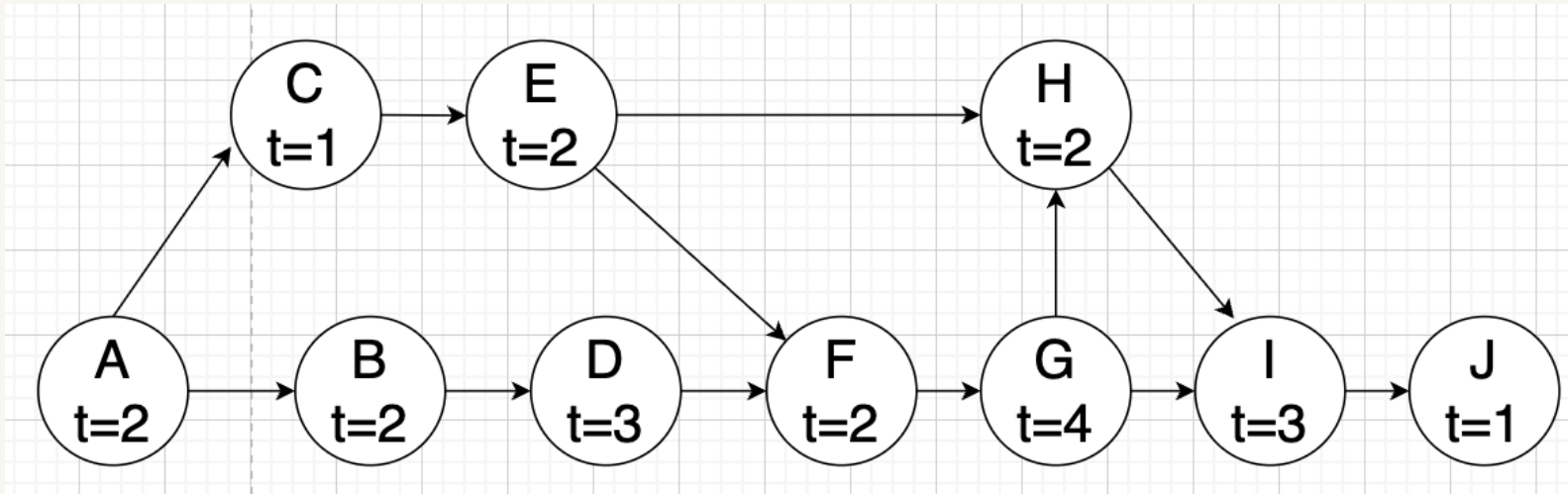
Activity Table

Activity #	Activity	Duration	IP
A	Collect	2	-
B	Analyze processes	2	A
C	Analyze data	1	A
D	Design processes	3	B
E	Design data	2	C
F	Design screens	2	D,E
G	Design reports	4	F
H	Program	2	E,G
I	Test and Document	3	G,H
J	Install	1	I

[illegible]

NETWORK DIAGRAM & CRITICAL PATH

Network Diagram



Critical Path

A → B → D → F → G → H → I → J
2 + 2 + 3 + 2 + 4 + 2 + 3 + 1 = 19 weeks

The network diagram shows how each task flows, and the critical path highlights the longest route through the project.

A series of thin, light-brown lines forming an abstract geometric pattern in the top-left corner of the slide. The lines intersect to create various triangular and polygonal shapes, some of which are nested within others.

PROJECT SUMMARY

This case study applied each phase of the Systems Development Life Cycle to design a modern Point of Sale system for a small retail business. I analyzed operational challenges, defined system requirements, created process and data models, and developed a complete project plan including timelines and critical path analysis. The proposed solution demonstrates how structured systems design can improve efficiency, organization, and customer experience.